

Lesson 9: Proof & Deductive Reasoning — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Write an expression for (a) an even number and (b) an odd number, using the integer n .
2. Write three consecutive integers, the smallest being n .
3. Write an expression for a multiple of 5.
4. Is the sum of two odd numbers odd or even? State your answer.
5. Give a counterexample to: 'every prime number is odd.'

Section 2 — Core practice

6. Prove that the sum of two even numbers is always even.
7. Prove that the sum of three consecutive integers is a multiple of 3.
8. Prove that the sum of an even number and an odd number is odd.
9. Disprove: ' $n^2 + n + 5$ is always prime.' (try $n = 4$)
10. Prove that the product of two even numbers is a multiple of 4.
11. Prove that $2n - 1$ is always odd for any integer n .

Section 3 — Challenge & exam-style

- 12.** Prove that the product of two odd numbers is odd.
- 13.** Prove that the sum of the first n odd numbers, $1+3+5+\dots+(2n-1)$, equals n^2 .
- 14.** Prove that the difference between the squares of two consecutive integers is always odd.
- 15.** Disprove: 'if n is an integer then $n^2 > n$.' (consider $n = 0$ or 1)